# FRACTIONS

*Short Revision Notes | Understand Level (Explain Ideas & Concepts)*

## 1. What is a Fraction?

| Concept | Explanation | Example |
|---|---|---|
| Definition | A fraction represents a part of a whole | 3/4 means 3 parts out of 4 equal parts |
| Whole | The complete object or quantity being divided | A circle, a collection of objects, a length |
| Equal Parts | The whole must be divided into EQUAL parts | A cake cut into 8 equal slices |
| Fraction Value | Shows how many parts are taken from the whole | 5/8 means 5 parts taken from 8 equal parts |

**Understanding the Concept:**

| Whole | Divided Into | Shaded | Fraction |
|---|---|---|---|
| Circle | 5 equal parts | 3 parts shaded | 3/5 |
| Team of 5 people | 5 members | 3 boys | 3/5 of team are boys |
| 100 rupees | 100 equal parts | 50 parts | 1/2 of 100 rupees = Rs 50 |

### Key Understanding

- The denominator tells you how many equal parts the whole is divided into.
- The numerator tells you how many of those parts you have.

## 2. Understanding Proper and Improper Fractions

| Concept | Explanation | Examples |
|---|---|---|
| Proper Fraction | Represents a value LESS than 1 | 1/2, 2/3, 3/4, 5/8, 7/8 |
| Why? | Numerator < Denominator (fewer parts than the whole) | 2 < 3, 3 < 4, 5 < 8 |
| Improper Fraction | Represents a value EQUAL TO or GREATER than 1 | 3/2, 5/3, 8/5, 11/8, 3/3 |
| Why? | Numerator >= Denominator (the whole or more) | 3 > 2, 5 > 3, 3 = 3 |

**Visual Understanding:**

| Fraction | Value | Explanation |
|---|---|---|
| 3/4 | Less than 1 | You have 3 parts out of 4 (less than a whole) |
| 4/4 | Equal to 1 | You have all 4 parts (exactly one whole) |
| 5/4 | Greater than 1 | You have 5 parts of size 1/4 (one whole + 1/4 more) |

**Understanding Mixed Numbers:**

- 1 1/2 means: 1 whole + 1/2 of another whole
- 2 3/4 means: 2 wholes + 3/4 of another whole
- 3 2/5 means: 3 wholes + 2/5 of another whole

```
1 1/2 =
[##########] [######....]
1 whole + 1/2
```

# 3. Understanding Equivalent Fractions

| Concept | Explanation | Example |
| --- | --- | --- |
| Definition | Different fractions that represent the SAME value | 1/2 = 2/4 = 3/6 = 4/8 |
| Why they're equal | They represent the same portion of the whole | 1/2 of a pizza = 2/4 of the same pizza |
| How to find them | Multiply or divide BOTH numerator and denominator by the SAME number | — |

```
1/2 = 2/4 = 3/6
[#...] [##..] [###...]
```

**Why It Works:**

| Fraction | Operation | Equivalent Fraction | Check |
| --- | --- | --- | --- |
| 1/2 | ×2/2 | 2/4 | 1×2=2, 2×2=4 |
| 1/3 | ×3/3 | 3/9 | 1×3=3, 3×3=9 |
| 3/4 | ×4/4 | 12/16 | 3×4=12, 4×4=16 |
| 4/6 | ÷2/2 | 2/3 | 4÷2=2, 6÷2=3 |
| 6/9 | ÷3/3 | 2/3 | 6÷3=2, 9÷3=3 |

### Key Understanding

- You are NOT changing the value — you are just dividing the whole into more or fewer equal parts.
- It's like using larger or smaller measuring cups — the amount of water stays the same!

## 4. Fractions, Decimals, and Percentages

| Concept | Fraction | Decimal | Percentage | Meaning |
|---|---|---|---|---|
| Half | 1/2 | 0.5 | 50% | 1 part out of 2 |
| Quarter | 1/4 | 0.25 | 25% | 1 part out of 4 |
| Three Quarters | 3/4 | 0.75 | 75% | 3 parts out of 4 |
| Tenth | 1/10 | 0.1 | 10% | 1 part out of 10 |

**How to Convert:**

| Fraction | Decimal | Percentage | How to Convert |
|---|---|---|---|
| 1/2 | 0.5 | 50% | 1 ÷ 2 = 0.5, 0.5 × 100 = 50% |
| 1/4 | 0.25 | 25% | 1 ÷ 4 = 0.25, 0.25 × 100 = 25% |
| 3/4 | 0.75 | 75% | 3 ÷ 4 = 0.75, 0.75 × 100 = 75% |
| 1/5 | 0.2 | 20% | 1 ÷ 5 = 0.2, 0.2 × 100 = 20% |

## 5. Understanding "OF" as Multiplication

| Concept | Explanation | Example |
|---|---|---|
| "of" in fractions | The word "of" means "multiply" | 1/2 of 100 = 1/2 × 100 = 50 |
| Why? | You are taking a portion OF the whole | 1/3 OF 20 kg = 1/3 × 20 kg = 6 2/3 kg |
| Fraction OF a Fraction | Multiplying fractions together | 1/2 OF 1/3 = 1/2 × 1/3 = 1/6 |

```
1/2 of 100 = 1/2 × 100 = 50
1/3 of 3/4 = 1/3 × 3/4 = 3/12 = 1/4
2/3 of 1 1/2 = 2/3 × 3/2 = 6/6 = 1
```

- When you see "of", think "multiply"!

# 6. Understanding Reciprocals

| Concept | Explanation | Example |
|---|---|---|
| Definition | Two numbers whose product is 1 | 2/3 and 3/2 -> 2/3 × 3/2 = 1 |
| How to find | Flip the numerator and denominator | Reciprocal of 3/4 = 4/3 |
| Why important | Used in division of fractions | 1/2 ÷ 1/3 = 1/2 × 3/1 = 1 1/2 |

| Number | Reciprocal | Product |
|---|---|---|
| 2/3 | 3/2 | 2/3 × 3/2 = 6/6 = 1 |
| 3/4 | 4/3 | 3/4 × 4/3 = 12/12 = 1 |
| 1/5 | 5/1 = 5 | 1/5 × 5 = 1 |
| 5 | 1/5 | 5 × 1/5 = 1 |
| 2 1/2 = 5/2 | 2/5 | 5/2 × 2/5 = 10/10 = 1 |

- Every number has a reciprocal except zero (because you cannot divide by zero).

# 7. Understanding Addition and Subtraction of Fractions

| Concept | Explanation | Example |
|---|---|---|
| Same Denominator | Add/subtract numerators, keep denominator | 1/5 + 3/5 = (1+3)/5 = 4/5 |
| Why? | The parts are the same size, so just count them | 1 fifth + 3 fifths = 4 fifths |
| Different Denominator | Must make them the SAME size first | 1/2 + 1/3 = 3/6 + 2/6 = 5/6 |
| Why? | Parts are different sizes, so need to convert | Halves and thirds are different sizes |

```
1/5 + 2/5 + 1/5 = ?
[#....] + [##...] + [#....] = [####.] = 4/5
5 equal parts: 1 + 2 + 1 = 4 parts -> 4/5
```

```
1/2 + 1/3 = ?
[#####.] [#####.] = 5/6

Convert to common denominator (6):
1/2 = 3/6 (divide whole into 6 parts, take 3)
1/3 = 2/6 (divide whole into 6 parts, take 2)
3/6 + 2/6 = 5/6
```

- You can only add or subtract fractions when they are the SAME TYPE of parts!

## 8. Understanding Multiplication of Fractions

| Concept | Explanation | Example |
|---|---|---|
| Rule | Multiply numerators, multiply denominators | 1/2 × 3/4 = 3/8 |
| Why? | You're finding a fraction of a fraction | 1/2 × 3/4 = 1/2 of 3/4 |
| Mixed numbers | Convert to improper fractions first | 1 1/2 × 2 3/4 = 3/2 × 11/4 |

```
1/2 × 3/4 means: Take 3/4 and find half of it

[######..] -> 3/4
Take 1/2 of it -> [###.....] -> 3/8
```

**Cancellation Concept:**

| Before Cancellation | After Cancellation | Understanding |
|---|---|---|
| 2/3 × 3/4 = 6/12 | 2/3 × 3/4 = 1/2 × 1/2 = 1/4 | Cancel 2 with 4, 3 with 3 |
| 3/4 × 8/9 = 24/36 | 3/4 × 8/9 = 1/3 × 2 = 2/3 | Cancel 3 with 9, 4 with 8 |

● Multiplication makes fractions SMALLER (when both are proper fractions).

## 9. Understanding Division of Fractions

| Concept | Explanation | Example |
|---|---|---|
| Rule | Multiply by the reciprocal of the second fraction | 3/4 ÷ 1/2 = 3/4 × 2/1 = 12/4 = 3 |
| Why? | Division asks "how many times does one quantity fit into another?" | How many 1/2s fit into 3/4? |
| Visual | 3/4 ÷ 1/2 = 1 1/2 | One and a half halves fit into 3/4 |

```
3/4 ÷ 1/2 means: How many 1/2s fit in 3/4?

[######..] = 3/4
[####....] = 1/2
[####....] + [##......] = 1 1/2 halves fit in 3/4
So 3/4 ÷ 1/2 = 1 1/2
```

**Understanding "Keep, Change, Flip":**

| Step | Example (1/2 ÷ 1/3) | Explanation |
|---|---|---|
| Keep | 1/2 | Keep the first fraction as is |
| Change | × | Change division to multiplication |
| Flip | 3/1 | Flip the second fraction (find reciprocal) |
| Calculate | 1/2 × 3/1 = 3/2 = 1 1/2 | Multiply the fractions |

- Division is just multiplication by the reciprocal.
- Dividing by a fraction makes the answer LARGER!

## 10. Understanding the BODMAS Rule

| Concept | Explanation | Example |
|---|---|---|
| B | Brackets (do inside first) | 2(3+1) = 2(4) = 8 |
| O | Of (multiplication) | 1/2 of 100 = 50 |
| D | Division (left to right) | 12 ÷ 4 = 3 |
| M | Multiplication (left to right after D) | 6 × 2 = 12 |
| A | Addition (left to right after M) | 5 + 3 = 8 |
| S | Subtraction (left to right after A) | 8 - 2 = 6 |

**Why Order Matters:**

| Expression | Without BODMAS | With BODMAS | Correct |
|---|---|---|---|
| 2 + 3 × 4 | 5 × 4 = 20 | 2 + 12 = 14 | 14 (correct) |
| 1/2 of 3 + 1 | 1/2 × 4 = 2 (left to right) | (1/2×3)+1 = 1 1/2+1 = 2 1/2 | 2 1/2 (correct) |
| 6 ÷ 2 × 3 | 6 ÷ 6 = 1 | 3 × 3 = 9 | 9 (correct) |

**Understanding "OF" in BODMAS:**

| Expression | BODMAS Order | Calculation |
|---|---|---|
| 1/2 of 20 + 3 | 1/2 × 20 + 3 | 10 + 3 = 13 |
| 1/2 × (20 + 3) | Brackets first | 1/2 × 23 = 11 1/2 |
| 2/3 of 3/4 × 1/2 | 2/3 × 3/4 × 1/2 (left to right) | (2/3×3/4)×1/2 = 1/2×1/2 = 1/4 |
| 2/3 ÷ 3/4 of 1/2 | 2/3 ÷ (3/4 × 1/2) = 2/3 ÷ 3/8 | 2/3 × 8/3 = 16/9 |

- BODMAS ensures everyone gets the SAME answer.
- Without it, multiplication and division would be interpreted differently.

# 11. Understanding Mixed Numbers

| Concept | Explanation | Example |
|---|---|---|
| Definition | A whole number plus a proper fraction | 1 1/2, 2 3/4, 3 2/5 |
| Representation | Shows more than one whole | 1 1/2 = 1 + 1/2 |
| Why needed | Easier to understand than improper fractions | 3/2 = 1 1/2 |

```
1 1/2 =
[##########] [######....]
1 whole + 1/2

2 3/4 =
[##########] [##########] [#######...]
1 whole 1 whole + 3/4
```

**Mixed to Improper Understanding:**

| Mixed Number | Process | Improper Fraction |
|---|---|---|
| 1 1/2 | 1 whole = 2/2, + 1/2 = 3/2 | 3/2 |
| 2 3/4 | 2 wholes = 8/4, + 3/4 = 11/4 | 11/4 |
| 3 2/5 | 3 wholes = 15/5, + 2/5 = 17/5 | 17/5 |

● A mixed number and an improper fraction are two ways of saying the SAME thing — they are EQUAL values.

## 12. Understanding the Relationship Between Parts

| Concept | Explanation | Fraction Example |
|---|---|---|
| Part of a Whole | The whole is divided into equal parts | 4/5 of the circle is shaded |
| Part of a Collection | The whole is a group of items | 3/5 of the team are boys |
| Part of an Amount | The whole is a quantity (money, weight, etc.) | 1/2 of 100 rupees = 50 rupees |
| Fraction of a Fraction | Taking part of a part | 1/2 of 1/3 = 1/6 |

```
A team has 5 people: 3 boys, 2 girls

Whole team = 5 people = 5/5
Boys = 3 people = 3/5 of the team
Girls = 2 people = 2/5 of the team

Boys + Girls = 3/5 + 2/5 = 5/5 = 1 (the whole team)
```

**Understanding Fractions of Collections:**

| Collection | Fraction | Number |
|---|---|---|
| 10 objects | 1/5 of 10 | 10 ÷ 5 = 2 objects |
| 30 students | 2/3 of 30 | 30 ÷ 3 × 2 = 20 students |
| 100 rupees | 3/4 of 100 | 100 ÷ 4 × 3 = 75 rupees |
| 20 kg | 3/5 of 20 | 20 ÷ 5 × 3 = 12 kg |

● "of" takes part of the whole — you divide by the denominator and multiply by the numerator.

## 13. Common Misunderstandings to Avoid

| Misunderstanding | Why It's Wrong | Correct Understanding |
|---|---|---|
| 1/2 + 1/3 = 2/5 | Can't add different sizes | Convert to same denominator: 3/6+2/6=5/6 |
| 1/2 × 1/2 = 1/4 | Actually correct! | 1×1=1, 2×2=4 (correct) |
| 1/2 ÷ 1/3 = 1/6 | Dividing by a smaller number makes LARGER | 1/2 × 3/1 = 1 1/2 |
| 3/3 = 0 | Not zero! | 3/3 = 1 (three thirds make a whole) |
| Improper fractions are “wrong” | They're just another way to write the same value | 5/3 = 1 2/3 |
| "OF" means addition | "OF" means multiplication | 1/2 of 20 = 10 (not 20 1/2) |

## 14. Understanding the LCM Concept

| Concept | Explanation | Example |
|---|---|---|
| LCM | Least Common Multiple | LCM of 3 and 4 is 12 |
| Why used | To find common denominator | 1/2 + 1/3 -> LCM(2,3) = 6 |
| Method | Find smallest number both denominators divide into | 2 divides into 6, 3 divides into 6 |

**Understanding With Examples:**

| Fractions | LCM | Convert | Result |
|---|---|---|---|
| 1/2 + 1/3 | 6 | 3/6 + 2/6 | 5/6 |
| 3/4 - 2/5 | 20 | 15/20 - 8/20 | 7/20 |
| 2/3 + 1/4 | 12 | 8/12 + 3/12 | 11/12 |
| 5/6 - 1/3 | 6 | 5/6 - 2/6 | 3/6 = 1/2 |

- The LCM becomes the "common language" that lets you compare and combine fractions.

## 15. Check Your Understanding

| No. | Question | Explain |
|---|---|---|
| 1 | What does the denominator tell you? | How many equal parts the whole is divided into |
| 2 | What does the numerator tell you? | How many parts you have taken |
| 3 | Why can't you add 1/2 + 1/3 directly? | The parts are different sizes |
| 4 | What does "of" mean in fraction problems? | Multiplication |
| 5 | What is a reciprocal? | Flipping the fraction upside down |
| 6 | Why is 5/4 greater than 1? | You have 5 quarters = 1 whole + 1 quarter |
| 7 | Why is 1/2 × 3/4 smaller than 1/2? | You're taking half of three quarters |
| 8 | Why is 1/2 ÷ 1/3 larger than 1/2? | Dividing by a fraction means "how many times does it fit?" |
| 9 | Why do equivalent fractions have different numbers? | The whole is divided into different numbers of equal parts |
| 10 | Why is BODMAS important? | So everyone performs operations in the same order |

## 16. Key Understandings to Remember

### * Master Summary

- FRACTIONS REPRESENT PARTS OF A WHOLE.
- THE DENOMINATOR TELLS THE NUMBER OF EQUAL PARTS. The numerator tells how many are taken.
- TO ADD/SUBTRACT, FRACTIONS MUST HAVE THE SAME DENOMINATOR.
- "OF" ALWAYS MEANS MULTIPLICATION.
- TO DIVIDE BY A FRACTION -> MULTIPLY BY ITS RECIPROCAL.
- MIXED NUMBERS AND IMPROPER FRACTIONS ARE EQUAL. They are just written differently.
- EQUIVALENT FRACTIONS HAVE THE SAME VALUE. They look different but represent the same amount.
- RECIPROCALS MULTIPLY TO GIVE 1.
- BODMAS TELLS YOU THE ORDER TO PERFORM OPERATIONS: Brackets -> Of -> Division -> Multiplication -> Addition -> Subtraction

***Revision Tip: Fractions are about understanding parts and wholes. Always visualize fractions to understand them — draw circles, bars, or use real objects like pizza or cake!***